

AML-REPORT

Cell 1

This cell installs the main libraries that we are going to use in the project like pandas, numpy, matplotlib and sklearn. These libraries help us in data analysis, visualization and applying machine learning models.

Cell 2

Here we import all the needed libraries into the notebook. This step is important so we can actually use the functions and tools from these libraries later on.

Cell 3

In this cell, the dataset is loaded from the UCI repository. It also displays the first few rows just to make sure that the data is loaded correctly and to get a first look at it.

Cell 4

This cell shows general information about the dataset such as number of rows, columns, and data types. It helps in understanding what kind of data we are dealing with before starting any processing.

Cell 5

Here we generate a statistical summary of the dataset. It shows things like mean, min, max and standard deviation. This gives us a better idea about how the values are distributed.

Cell 6

This cell checks if there are any missing values in the dataset. It is important because missing values can affect the results if not handled properly.

Cell 7

In this cell we check for duplicate rows. Duplicate data can sometimes bias the model so it’s good to make sure there are none.

Cell 8

This cell selects only the numerical features related to customer spending. We remove Channel and Region because they are categorical and not suitable for clustering directly.

Cell 9

Here we create histograms for each feature. This helps us see the distribution of the data and whether it is skewed or not.

Cell 10

This cell creates boxplots for all features. The goal is to visually detect outliers and understand how spread out the data is.

Cell 11

In this step, we calculate Z-scores to detect outliers in a more numerical way. It helps confirm what we saw in the boxplots.

Cell 12

This cell saves a copy of the original data before scaling. This is useful later when we want to interpret the clusters using real values.

Cell 13

Here we apply standardization using StandardScaler. This is important because the features have very different ranges and we want them to contribute equally.

Cell 14

This cell shows histograms after scaling. It helps confirm that the scaling worked correctly.

Cell 15

In this cell, we calculate and display the correlation matrix. This helps us understand relationships between features, like which variables are strongly related.

Cell 16

This cell creates pairplots between features. It gives a more detailed visual view of relationships between variables.

Cell 17

Here we apply PCA to reduce the dimensionality of the dataset. This is mainly done for visualization purposes.

Cell 18

This cell plots the PCA results in 2D. It helps us see if there are any visible groups or patterns in the data.

Cell 19

In this cell, we apply t-SNE which is another dimensionality reduction technique but non-linear. It can sometimes show clusters better than PCA.

Cell 20

This cell visualizes the t-SNE output. It gives another perspective on how the data might be grouped.

Cell 21

Here we import clustering algorithms and evaluation metrics. These will be used later to build and evaluate the models.

Cell 22

This cell applies the Elbow Method using K-Means for different values of K. It helps us decide the possible number of clusters.

Cell 23

In this cell, we plot the Elbow curve. We try to find the point where the curve starts to flatten.

Cell 24

This cell defines a function to calculate the Dunn Index. It is used to measure how well clusters are separated.

Cell 25

Here we define the Gap Statistic function. It compares our clustering with random data to see if the clusters are meaningful or just random.

Cell 26

This cell applies K-Means clustering for K = 2, 3, and 4. The results are stored for later use.

Cell 27

Here we visualize the K-Means results using PCA. It helps us see how the clusters look.

Cell 28

This cell applies Spectral Clustering for the same K values. It is another clustering method that can capture more complex structures.

Cell 29

This cell visualizes the Spectral Clustering results. It allows us to compare visually with K-Means.

Cell 30

Here we apply Gaussian Mixture Model (GMM). It is a probabilistic clustering method.

Cell 31

This cell visualizes the GMM results. It helps us compare all models visually.

Cell 32

This cell evaluates all clustering results using different metrics like Silhouette Score, DBI, CH, Dunn Index and Gap Statistic. This is a very important step to compare models.

Cell 33

Here we format the evaluation results to make them easier to read.

Cell 34

This cell identifies the best model for each metric individually. It shows which model performs best in each case.

Cell 35

This cell creates bar charts for the metrics. It helps visualize the differences between models.

Cell 36

Here we define a function for cluster profiling. It calculates the average values for each cluster using the original data.

Cell 37

This cell applies profiling for K-Means clusters. It helps us understand what each cluster represents.

Cell 38

Here we do the same profiling but for Spectral Clustering.

Cell 39

This cell profiles the GMM clusters. Now we have interpretation for all models.

Cell 40

This cell automatically selects the best model based on multiple metrics. This is better than choosing manually.

Cell 41

Here we retrieve the cluster profile of the selected best model.

Cell 42

This cell visualizes the cluster profile using a heatmap. It makes it easier to compare clusters.

Cell 43

This cell sorts the evaluation results to show the best models first.

Cell 44

Finally, this cell prints the final conclusion, including which model and K value were selected and why.

📌 Overall Summary

In this project, we performed clustering on the Wholesale Customers dataset to group customers based on their spending behavior. First, we cleaned and explored the data, then we scaled it to make sure all features have equal importance. After that, we used PCA and t-SNE just for visualization to understand the data better.

Then, we applied three clustering algorithms: K-Means, Spectral Clustering and GMM, using different values of K. We evaluated all models using multiple metrics instead of relying on just one, which makes the results more reliable.

Finally, we selected the best model automatically and analyzed the clusters using the original data to understand what each group represents. Overall, the process gives a complete pipeline from raw data to meaningful customer segmentation.

📌 Conclusion

The project shows that clustering can be used to identify different types of customers based on their spending. By comparing multiple models and evaluation metrics, we were able to choose the best clustering solution in a more objective way. The final clusters provide useful insights into customer behavior, which can be helpful for business decisions.
